

Characteristic Analysis of Subsynchronous Resonance in Practical Wind Farms Connected to Series-Compensated Transmissions

Xiaorong Xie, *Senior Member, IEEE*, Xu Zhang, Huakun Liu, Hui Liu, Yunhong Li, and Chuanyu Zhang

Abstract—The emerging subsynchronous resonance (SSR) caused by the interaction of wind turbine generators (WTGs) with series compensation has aroused great concerns. For this particular issue, this paper is aimed to fill the gap between theoretical studies and actual observations. By analyzing the field data of 58 SSR events captured in a practical wind power system and examining the observed dynamics with previous theoretical results, the mechanism and characteristics of SSR are revealed in a more explicit and substantial way. The necessary conditions and dominant influential factors are identified and the underlying reasons are discovered. Theoretically derived as well as practically measured impedance models have demonstrated that the converter control of doubly fed induction generator (DFIG) produces negative resistance at the slip frequency and thus causes unstable SSR; while permanent magnet synchronous generators and self-excited induction generators are just passively engaged in those SSR incidents. The distribution of the oscillation frequency has also been examined with field measurements. It is discovered that WTGs at different locations participate into the same SSR mode and their frequencies are not fixed but keep changing with the time, the variation of grid topology, and the number of online generators.

Index Terms—Doubly fed induction generator (DFIG), permanent magnet synchronous generator (PMSG), series compensation, subsynchronous control interaction (SSCI), subsynchronous resonance (SSR).

I. INTRODUCTION

WITH the increasing applications of wind power energy and series compensation, there is a growing risk of subsynchronous resonance (SSR) caused by the interaction between doubly fed induction generators (DFIGs) and series-compensated transmissions [1], [2]. This emerging type of SSR,

also known as subsynchronous control interaction (SSCI), has been detected worldwide, for instance, in the wind farms of USA (ERCOT and Xcel Energy) and China (Guyuan, Hebei Province) [3]–[5]. As it would seriously weaken the safe and reliable operation of the wind farms as well as the whole transmission system, the issue has aroused great concerns from both academic and industrial communities.

Previously, theoretical research and simulation studies have been conducted to investigate the mechanism and characteristics of this SSR phenomenon [6]–[28]. Several control schemes have also been proposed to mitigate the problem [29]–[37]. However, there is still a huge gap between theoretical analyses and actual observations. Most importantly, existing studies have only revealed the SSCI-type SSR in very simple system scenario, for instance, considering one wind farm with an aggregated wind turbine generator (WTG) or multiple identical ones connected to a common bus. But in practice, there would be hundreds or even thousands of WTGs with diverse types or parameters and they are widely deployed at different locations. What is the characteristics of SSR in such a real scenario? Specifically, which and how WTGs of different types and at different places participate into the subsynchronous interaction. These are questions to be properly addressed with real data of actual SSR events. Another important issue is about the mutual verification between the theory and practice. Many key theoretical results regarding the mechanism and features of the SSR have been provided based on mathematical models. For example, the negative-resistance effect brought by DFIG's converter control is essential to the occurrence of SSR. And its stability is determined by the dominant factors such as wind speed and compensation degree. However, these theoretical outputs have not been fully verified by empirical evidence or in a practical system.

To address these issues, the paper will present a detailed characteristics analysis of the emerging SSR based on real-world data gathered from actual SSR events that have been captured in a large-scale wind farm system interfaced with series-compensated transmissions, i.e., Guyuan wind power system located in North China, Hebei Province. New materials to be provided in the paper are in the following aspects:

- 1) For the first time, the characteristics of SSCI-type SSR are investigated for a practical large-scale wind-farm system, containing more than 3000 WTGs and four series-compensated lines. The participation of three types of

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X. Xie, X. Zhang, H. Liu, and C. Zhang are with the State Key Laboratory of Power System, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China (e-mail: xiexr@tsinghua.edu.cn; zhangxu2016@tsinghua.edu.cn; goodliu@huakun@163.com; zhangcy10@foxmail.com).

H. Liu and Y. Li are with the North China Electric Power Research Institute, Beijing 100045, China (e-mail: liu.hui@ncepri.com.cn; 13260208572@163.com).

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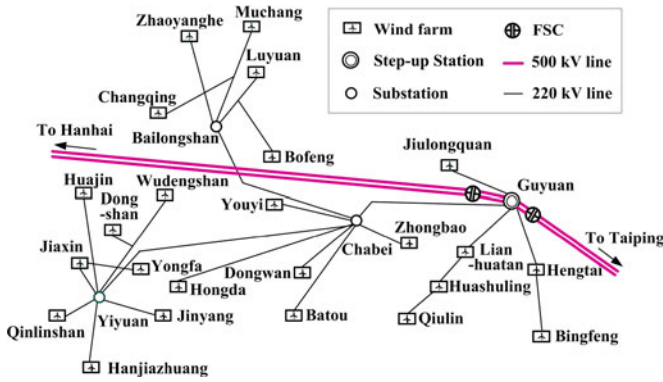


Fig. 1. Single-line diagram of the wind power system in Guyuan area.

WTGs, namely DFIG, permanent magnet synchronous generator (PMSG) and self-excited induction generators (SEIG), are revealed based on field measurements. The impacts from the diversity of location, wind speed and operating condition will also be explored. The characteristics summarized from 58 SSR events are to be compared with those provided by theoretical studies.

- 2) The characteristics of oscillatory frequency, for instance, its temporal and geographic distribution in a wide-range of SSR events, are very important for the physical understanding of such new type of SSR. However, this has not been discussed in previous work. In this paper, the wide-area measurements are collected and synchronized to offer a global view of the oscillation. It is revealed that during an SSR event the resonant frequency is time-varying but consistent at different locations.
- 3) The frequency-dependent impedance models (IMs) are identified with field data for both DFIGs and PMSGs. The negative-resistance effect of DFIG in the SSCI is validated. In contrast, PMSGs exhibit as high impedances, which confirms their passiveness in the discussed SSR events.
- 4) The SSR characteristics observed in the target Guyuan system are also compared with those discovered in other systems, such as ERCOT and Xcel Energy, and the striking differences are summarized and presented.

The organization of the paper is as follows. Section II describes the Guyuan wind power system and its SSR events. Section III examines the various distinct features of the SSR captured at Guyuan and compares them with those happening elsewhere in the world. In Section IV the field-measured IMs are presented and used to verify actual observations with previous theoretical results. Section V concludes this paper.

II. THE WIND POWER SYSTEM AND ITS SSR EVENTS

A. The Wind Power System

The wind power system in the Guyuan area is located in the northwest of Hebei Province, China. As show in Fig. 1, it has twenty-four wind farms by the end of 2014. All these wind farms distributed throughout the Guyuan area are radially connected to the 220 kV substations of Guyuan, Chabei, Yiyuan and

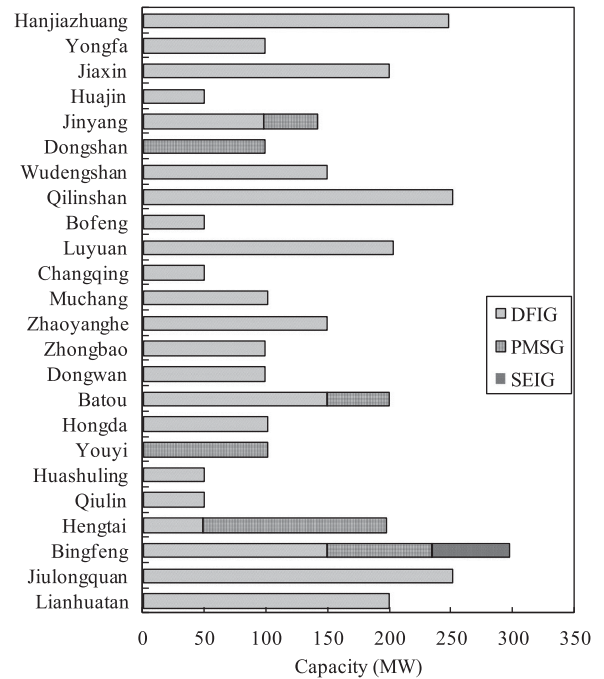


Fig. 2. The types and capacities of wind farms in Guyuan area.

Bailongshan. Then, the electric power is collected at the 500 kV Guyuan substation and next transmitted along two 500 kV corridors, each with double-circuit series-compensated transmission lines, to the North-China power grid. Four sets of fixed series capacitors (FSCs) are installed on the two parallel transmissions connecting Guyuan substation with Hanhai and Taiping substations. And the compensation degrees are 40% and 45%, respectively.

There are three types of WTGs, namely SEIG (Type 2), DFIG (Type 3) and PMSG (Type 4), installed at those wind farms. Among the total 3426.55 MW installed capacity, 82.8%, 15.4% and 1.8% of the wind power are provided by DFIGs, PMSGs and SEIGs, respectively. Clearly, DFIG accounts for most of the proportion. For each wind farm, the installed capacity and the types of WTG are illustrated in Fig. 2. Most wind farms contain only single type of WTG. For example, the wind farms of Hanjia Zhuang, Zhongbao, Muchang and Qilinshan are entirely made up of DFIGs; while Youyi and Dongshan are purely PMSG-based. A very few farms, for instance, Jinyang, Batou, Hengtai and Bingfeng, contains two or three types of WTGs.

B. SSR Events

In the target system, abnormal electrical oscillations had been observed since the commission of series capacitors in 2010. As the occurrence was fairly rare and had no obvious impact on the normal operation of the system, those oscillations did not cause sufficient attention initially. However, with more and more new wind farms put into operation in the area, SSR began to occur frequently. From December 2012 to December 2013, as many as 58 SSR events were detected. Such severe SSR would inevitably disturb the safe operation of the system. The immediate consequence is that many (even over one thousand

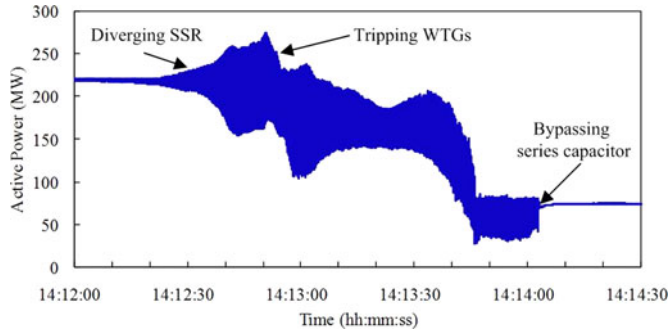


Fig. 3. The power flow of the step-up transformer during the SSR event.

in a case) WTGs were tripped due to the excessive current caused by the divergent SSR, which resulted in an abrupt and large decrease of wind power. SSR would also endanger the performance of power equipment. For instance, it causes large inter-harmonic currents flowing into transformers, which then vibrates abnormally and makes unusually loud noise. As a result, the aging of insulation would be sped up and the loss would be increased. Therefore, SSR is regarded as a big threat to the safety and stability of the whole system and calls for in-depth studies.

To illustrate the phenomena, a typical SSR event is taken as example. It happened on March 19th, 2013. Fig. 3 displays the active power flowing through the step-up transformer at Guyuan substation. The data were recorded by the wide-area monitoring system or WAMS. Initially, the wind power is relatively low, about 13% of the total capacity. At 14:12:15, the power started to oscillate in a divergent way. The oscillatory magnitude increased rapidly. At 14:12:48, the amplitude of oscillation reached to about 25% of the average power. As a result, many WTGs dropped out due to the intolerable inter-harmonic current. As the number of online WTGs became less, the average power decreased steadily. However, SSR still existed. Not until 14:14:03, when one of the series capacitors was bypassed, did SSR cease. The oscillation lasted for 108 seconds. The wind power decreased from 219.5 MW to 74.5 MW, causing a loss of 66% of the initial output.

Segmental curves of voltage and current of the transformer at the 220 kV side are displayed in Figs. 4 and 5. As shown in Fig. 4(a), the current is distorted seriously. The spectrum of the current is presented in Fig. 4(b), in which the subsynchronous frequency is observed to be about 8.1 Hz. The amplitude of the subsynchronous current is larger than half of the fundamental one. There is also a super-synchronous component at 91.9 Hz in the current. But it is much smaller. In contrast, the voltage, as shown in Fig. 5(a), is not so much distorted. Its spectrum in Fig. 5(b) confirms this observation. Although there are visible components of 8.1 Hz and 91.9 Hz in the voltage, they are almost negligible compared with the fundamental voltage.

III. CHARACTERISTIC ANALYSIS OF THE SSR

A. The Necessary Conditions for SSR Occurrence

It is an important issue that under what conditions SSR would happen. To find the answer, the different scenarios of the 58 SSR

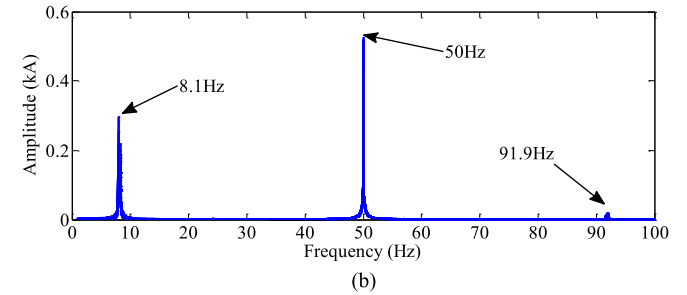
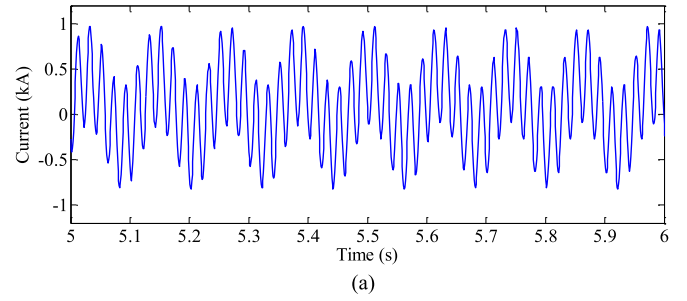


Fig. 4. The current during a typical SSR event. (a) Instantaneous current. (b) Spectrum of the current.

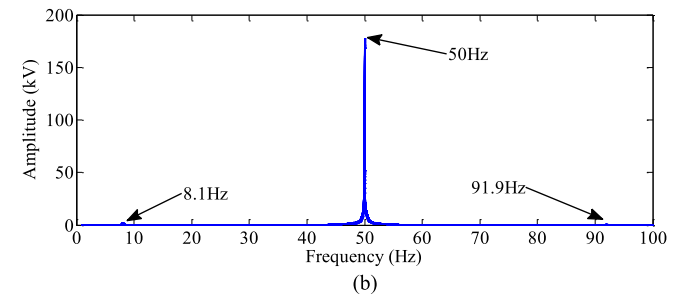
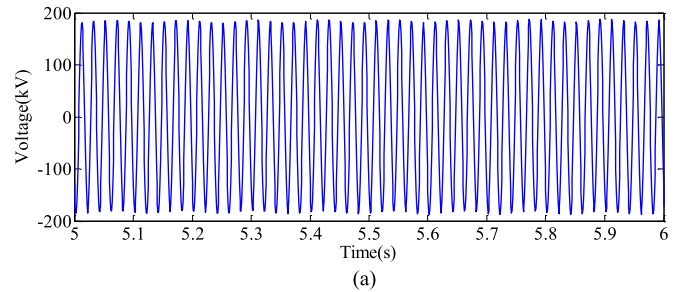


Fig. 5. The voltage during a typical SSR event. (a) Instantaneous voltage. (b) Spectrum of the voltage.

events were examined carefully, intending to get clues that led to the oscillations. Several factors, including the state of series capacitors, the wind speed and/or output level of wind power, are discovered to determine the occurrence of SSR.

According to the recorded data, all SSR events happened when the four sets of series capacitors were in service. If one or more series compensation are out of service or deliberately bypassed, SSR would never occur or be suppressed. Actually, about 90% cases of dangerous SSR were terminated by timely quitting one of the four sets of series capacitors. Obviously, the existence of all series compensation is a prerequisite for the

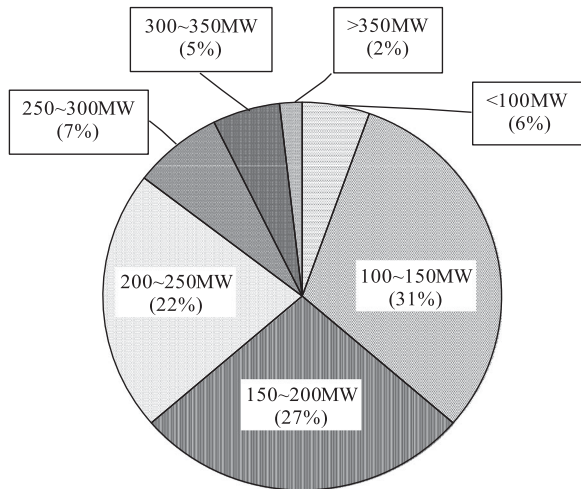


Fig. 6. The number and percentage of SSR events versus the wind power.

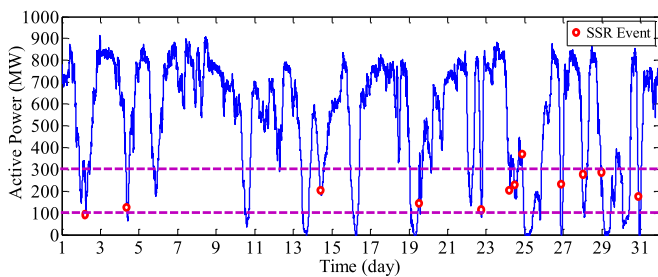


Fig. 7. The power flow through Guyuan Step-up station in March 2013.

occurrence of SSCI-type SSR. This observation is consistent with the theoretical results stated in [1].

Another pronounced observation is that when SSR occurred, the output level of wind power was usually small. Among the 58 SSR events, the average wind power output was in the range from 88 MW to 360 MW, or 2.6% to 10.5% of the total installed capacity. The number and percentage of SSR events versus the wind power are illustrated in Fig. 6. It can be seen that most (about 87%) of the captured SSR incidents happened when the average wind power is between 100 MW and 300 MW.

Specifically, Fig. 7 displays the upstream wind power in March 2013. During that period, 12 SSR incidents were recorded and their initial power was also marked in the figure.

Similarly, most (10 out of 12) SSR events appeared when the average wind power is between 100 MW and 300 MW. What calls for special attention is that SSR started when the power is in a downward trend. In other words, its starting point is on the left side of the power valley. Noticeably, there are some sharp downward trends (below 100 MW) where the SSR event did not occur. This is due to a low limit preset by the system operators. Before 2012, when the installed capacity of wind power was much lower, it was discovered that SSR appeared as the power was less than 100 MW. Then, to avoid SSR, the system operators set a low limit (100 MW) for the wind power, which, if violated for a few seconds, one set of series capacitors would be bypassed. As a result, for these occasions when wind power dropped quickly below that limit, no SSR was captured

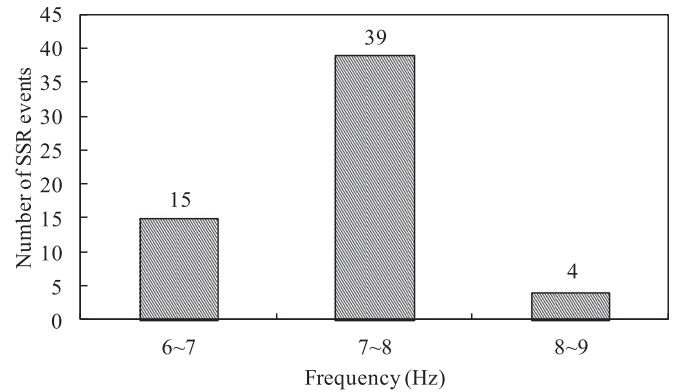


Fig. 8. The distribution of SSR frequency.

due to the immediate switch-off of series capacitors. So in Fig. 7, some of the sharper downward power valleys had no marks for SSR incidents. Of course, this previously set limit was proved to be unable to avoid SSR completely.

According to the statistic data provided by North-China Grid, the in-service rate of WTGs in this area is generally about 80% or less. Hence, 100 MW and 300 MW correspond to about 4% and 11% of the nominal capacity. In the light of the wind-speed-versus-output-power characteristics of the actual WTGs, it can be derived that unstable SSR happens when the average wind speed is in the rough range of 3 to 5 meters per second. It is a relatively lower speed. This observation about the influence of wind speed on SSR is consistent with the analysis in [1], [5], [6], [31], [34]. However, further decrease of wind speed leads to the dropping out of more WTGs. Considering the impact from the number of online WTGs [5], the probability of SSR would be reduced instead.

From the above analyses, it can be concluded that there are two necessary conditions for an unstable SSR incident. They are: 1) series compensation adopted in the transmission of wind power predominantly generated by DFIGs, and 2) a relatively low output level or wind speed. Therefore, for DFIG-based wind farms interfaced with series-compensated transmissions, even when the power system itself suffers no evident disturbance, a sudden or steady decrease of wind speed or in turn power output would trigger an unstable SSR incident. These conclusions are very consistent with the previous theoretical analyses [6], [31].

B. The Frequency Characteristics of the SSR

The frequency characteristics are very important in analyzing the features of SSR, including both pure electric resonance and torsional interaction. The distribution of the frequency at the start time of the mentioned 58 SSR events is illustrated in Fig. 8. Obviously, the oscillatory frequency is in the range from 6 to 9 Hz. And 54 out of the 58 SSR events have their frequencies varying from 6 to 8 Hz, accounting for 93.1%. However, the natural frequency of the WTG shaft is about 1.8 Hz and its complementary frequency is 48.2 Hz [1]. The frequency of the captured SSR is far from the torsional frequency. Therefore, this SSR can be ruled out from the traditional torsional interaction.

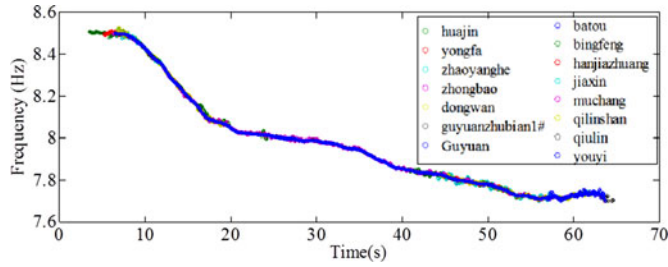


Fig. 9. The spatial and temporal distribution of the resonant frequency.

It is a different and new type of SSR without the participation of the mechanical dynamics of the turbine shaft.

Another evidence is that the frequency of this new SSR is not constant in even a single event. To demonstrate this, the frequencies recorded at 15 different wind farms that experienced the same SSR event are presented in Fig. 9. Starting at around 8.5 Hz, when the oscillation become intense enough that some WTGs were tripped about 3 seconds later. As a result, the frequency began to decrease. As the SSR continued, more and more WTGs dropped out and the oscillatory frequency declined steadily from 8.5 Hz to 8.1 Hz in 10 seconds and then to 7.7 Hz in 30 seconds. Despite the continuous change or decrease of the frequency with time, the frequency at different locations is perfectly identical, which means that all WTGs and the network are engaged in the same SSR mode.

As pointed out in our previous work [31], [34], the oscillation frequency of this new type of SSR is mainly determined by the equivalent inductance and capacitance, which involves the number/parameters of both transmissions and WTGs. When the number of grid-connected WTGs decreases due to the protective tripping action, the equivalent reactance of the wind farms increases, resulting a decline of oscillatory frequency. So the observations from Fig. 9 agree very well with existing theoretical results [31].

C. Participation of DFIGs and PMSGs Into the SSR

The vast majority (98.2%) of the WTGs in Guyuan area are DFIGs and PMSGs. Massive data are collected and analyzed to examine the participating degree of these two widely used WTGs into the SSR dynamics.

Typical curves are shown in Fig. 10, where the active power outputs of a DFIG-based wind farm, namely Hongda, and a PMSG-based wind farm, namely Youyi, are depicted during an SSR event that started at 14:09:00, March 19, 2013.

Except the diverse types of WTGs, both wind farms have 67 generators (each of 1.5 MW) and their total capacities are 100.5 MW. The initial power outputs are about 16% and 22% for Hongda and Youyi, respectively. Then there was probably a decrease of wind speed that triggered the SSR incident. As the oscillation built up, the amplitude of the oscillatory power at Hongda grew rapidly, with its maximum value reaching nearly 50% of the average power. Consequently, a lot of DFIGs were tripped and the output power fell quickly. When the oscillation damped out some 2 minutes later, only about 38% of the initial power was preserved. However, PMSGs at Youyi performed

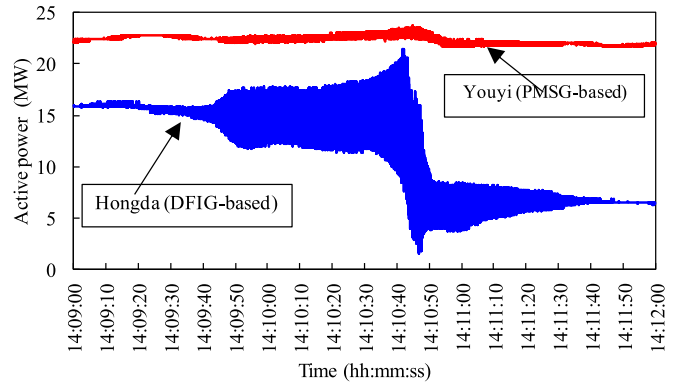


Fig. 10. The wind power output of Youyi and Hongda during an SSR incident.

quite differently and the peak value of oscillatory power was less than 5%. No PMSG dropped out in the event and the wind power remained almost the same when the SSR disappeared.

By comparing the response of the wind farms at Hongda and Youyi, it can be concluded that DFIGs play an incentive role in the SSR event. On the contrary, PMSG behaves passively. This observation is consistent with the theoretical results that [16]: Type 3 WTGs tend to provide negative damping for such type of SSR; while Type 4 WTGs make no contribution. Such distinct response characteristics will be further interpreted in the following section with the concept of IM.

D. Comparison With the SSR in ERCOT and Xcel Energy

Similar SSR incidents have been reported for the wind farms at Guyuan, ERCOT and Xcel Energy. They were all caused by the interaction between DFIGs and series-compensated transmissions. However, there are still striking differences between the characteristics of SSR captured at Guyuan and those in ERCOT and Xcel Energy.

First of all, the level of series compensation is different. It was reported that the transmission lines in ERCOT and Xcel Energy are respectively 75% and 60% series-compensated at the moment of SSR occurrence [3], [4]. However, the compensation levels in the studied Guyuan system are 40% and 45% for the lines. Essentially, the total equivalent compensation degree (TECD), which is defined as the ratio of capacitive and reactive impedances viewed from the wind farms, determines the characteristics of SSR. For the Guyuan system, the TECD is only 6~7% [5]. This very low value results in a much smaller oscillatory frequency (6–8 Hz) than those of ERCOT and Xcel Energy, which are 22 Hz and 13 Hz, respectively [3], [4]. Moreover, the incidents at Guyuan shows that such kind of SSR would happen even under the context of very low TECD. This is a very important reminder for the planners and operators of similar systems.

In the system of either ERCOT or Xcel Energy, the TECD is relatively low when the wind farms are connected to the meshed networks. Thus SSR does not appear during normal situations. Only after fault-induced or experimental trip of lines, were the DFIG-based wind farms suddenly switched to the connection with the system in a radial way through series-compensated

transmissions [3], [4]. Consequently, SSR incidents were triggered abruptly. However, fault or switching operation is not the prerequisite for the SSR incidents at Guyuan. Unstable SSR would be gradually self-excited just when the wind speed is entering an unfavorable range, for instance, from 3 to 5 meters per second. Therefore, the SSR incidents of ERCOT & Xcel were more abrupt than those of Guyuan. But SSR at Guyuan occurred more frequently and less predictably.

As far as the harmfulness and countermeasures are concerned, there would be quite different consideration for the different cases. For the cases of ERCOT and Xcel Energy, severe SSR would occur suddenly following a fault or switching action, the resulted excessive current inrushes into the WTGs and make their crowbars act immediately. Such large current/voltage stress would even exceed the design limits and thus damage the crowbar circuits or at worst the generators. Therefore, in these cases the WTGs or series capacitors should be tripped as fast as possible by a certain protection relay. But for the Guyuan system, SSR builds up gradually and would not cause immediate damage to the WTGs. So there is enough time for a feedback control to react to damp the oscillation. Protection relays are used only as backup countermeasures.

IV. IMPEDANCE MODEL BASED ANALYSES

Impedance model (IM) is widely used for the stability analysis of power systems incorporating power electronic converters [39]. This is also the case for the SSCI-type SSR due to the involvement of the converter controls of DFIGs and/or PMSGs. It has been theoretically proved that the DFIG-associated SSR existing in a series-compensated system is a particular type of control interaction with the negative damping or resistance mainly contributed by the rotor-side converter (RSC) of DFIG [5]. So if IM is available for a WTG or wind farm, the characteristics of SSR, including the frequency and damping, could be derived with many mature methods, for instance, the Nyquist stability criterion [38], [39]. Although the impedance-based theory is well established, there is still a lack of empirical evidence for the impedance characteristics of practical wind generators or farms. In this section, the theoretically derived IMs of the above-mentioned three types of WTGs will be reviewed and their field-measured results will be presented. The cross validation of theoretical and measured impedances further deepens the understanding of the characteristics of the SSR.

A. The IMs of DFIG, PMSG and SEIG

For a radial system with series capacitors in the main grid and WTGs at the sending terminals, both the grid and the wind farms can be simply represented with IMs for the purpose of analyzing the SSR characteristics [38]. As shown in Fig. 11, the grid is simplified into an R-L-C circuit; while the wind farms based on DFIG, PMSG or SEIG can be equivalently replaced by resistance in series with inductance or capacitance. If the negative resistance generated by the wind farms overcome the positive resistance of the grid, unstable SSR would happen and the frequency of oscillation is determined by the total inductance and capacitance [39].

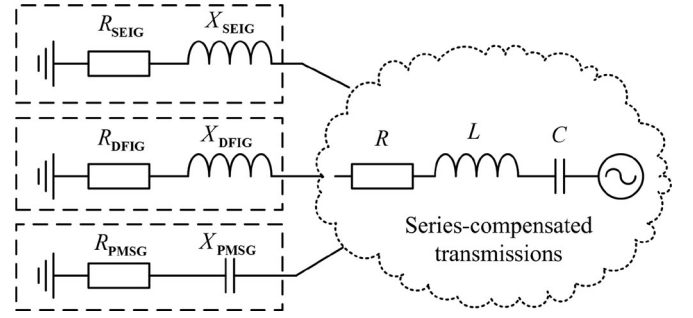


Fig. 11. The IMs of a radial system collecting wind power through series-compensated transmissions.

SEIG has no power electronic converter. Its IM is relatively simple, as expressed by:

$$\begin{aligned} Z_{SEIG} &= (R_{SEIG} + jX_{SEIG})/n \\ &= \{(r_r/s + j\omega_{SSR}L_r)/(j\omega_{SSR}L_m) \\ &\quad + R_s + j\omega_{SSR}L_s\}/n \end{aligned} \quad (1)$$

where R_{SEIG} (X_{SEIG}) is the equivalent resistance (reactance) of the SEIG-based wind farm; r_r (L_r) is the rotor resistance (inductance); R_s (L_s) is the stator resistance (inductance); L_m is the excitation inductance; ω_{SSR} is the angular frequency of SSR; ω_r is the rotor speed; and $s = (\omega_{SSR} - \omega_r)/\omega_{SSR}$, standing for the slip between ω_{SSR} and ω_r ; n is the number of the same type of SEIGs in the wind farm.

The IM of a wind farm containing n identical DFIGs can be expressed by [5], [40]:

$$\begin{aligned} Z_{DFIG} &= (R_{DFIG} + jX_{DFIG})/n \\ &= \{[(K_p + r_r)/s + j\omega_{SSR}L_r]/(j\omega_{SSR}L_m) \\ &\quad + R_s + j\omega_{SSR}L_s\}/n \end{aligned} \quad (2)$$

where R_{DFIG} (X_{DFIG}) is the equivalent resistance (reactance) of the DFIG-based wind farm; K_p is the proportional gain of the RSC control.

As the slip s is generally negative at the SSR frequency, the first items in both (1) and (2) present negative resistance. Noticeably, in (2) there is an additional item or K_p/s caused by the proportional gain (K_p) of the RSC. It would significantly increase the absolute value of the negative resistance and thus contribute a lot to an unstable oscillation [4]. Consequently, DFIG is made much more vulnerable to the SSCI-type SSR.

PMSGs are connected to the grid through full-scale voltage-sourced converters (VSCs). So its IM is quite different from those (Types 1–3) WTGs based on induction machines. Extensive work has been done to get its analytic expressions [16], [38]. The IM of PMSG is widely recognized to be jointly determined by schemes/parameters of its inner/outer-loop controllers, phase-locked-loop (PLL), dc dynamic and interfacing filters. Generally, a wind farm with n identical PMSGs can be expressed by:

$$Z_{PMSG} = (R_{PMSG} + jX_{PMSG})/n \quad (3)$$

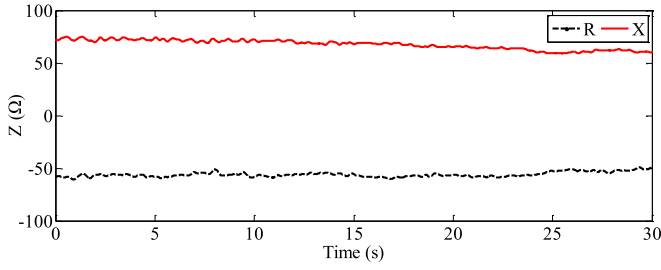


Fig. 12. Measured impedance of Hanjiazhuang (DFIG-based).

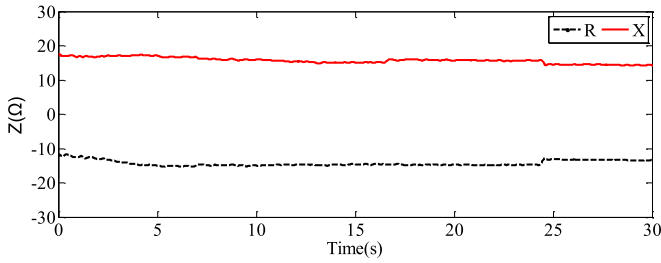


Fig. 13. Measured impedance of Zhongbao (DFIG-based).

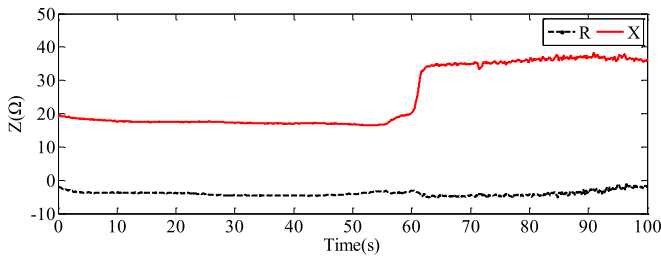


Fig. 14. Measured impedance of Jiulongquan (DFIG-based).

According to [16], at low subsynchronous frequency the equivalent resistance R_{PMSG} is usually a large and positive number and the equivalent impedance X_{PMSG} is a large but negative (capacitive) number.

Based on the theoretically derived IMs (1)–(3), the total impedance of the system can be obtained, which is next used to determine the damping (or stability) and frequency of SSR. It is widely believed that the negative-resistance effect of DFIG makes it the primary cause for this SSCI-type SSR while SEIG and PMSG usually play passive roles. However, such belief needs to be verified with field data.

B. Measured Impedances

To verify the concept of IMs, we identify their real values from field measured data by using the following formula:

$$Z_{\text{SSR}} = \dot{U}_{\text{SSR}} / \dot{I}_{\text{SSR}} \quad (4)$$

where $\dot{U}_{\text{SSR}}$ and $\dot{I}_{\text{SSR}}$ are subsynchronous voltage and current phasors extracted from the recorded data.

For illustration, the IMs of wind farms are identified using the data of the SSR event on March 19, 2013. Some typical results are shown in Figs. 12–16.

The IMs of Hanjiazhuang (Fig. 12), Zhongbao (Fig. 13) and Jiulongquan (Fig. 14) have confirmed that a DFIG-based

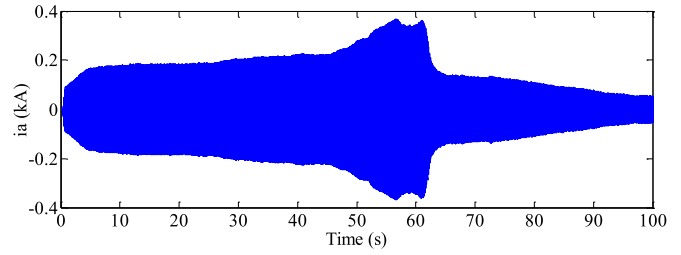


Fig. 15. The Measured current output of Jiulongquan.

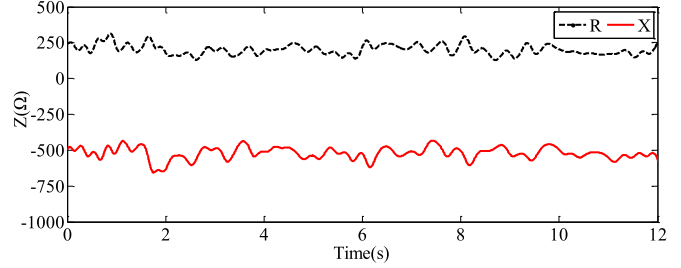


Fig. 16. Measured impedance of Youyi (PMSG-based).

wind farm appears as negative resistance and positive reactance. Fig. 16 displays the IM of a PMSG-based wind farm (Youyi). It is characterized by positive resistance and capacitive reactance, both of which have much large absolute values than those of DFIG-based wind farms. This has well confirmed PMSG's passiveness in the SSR dynamics.

The variation of the impedance over time helps to judge whether there are obvious changes in wind speed and/or number of online WTGs during the incident. For instance, Fig. 14 indicates a step change in the reactance of Jiulongquan. It was caused by the drop-out of DFIGs in that wind farms due to the excessive oscillatory current, which could be seen in Fig. 15.

C. IM-Based Characteristic Analysis

By comparing the theoretically derived IMs with those field-measured ones, we could conclude that they are consistent in general. With the IM concept confirmed, the characteristics of the discussed SSR could be interpreted in a more explicit and substantial way.

1) *Impedance Network of the System*: As the WTGs and the grid represented by IM, the system shown in Fig. 1 can be configured as an impedance network. By means of series and parallel connection, such a radial electric network is then easily reduced to a lumped impedance, namely Z_{Σ} , as shown in Fig. 17. By analyzing the frequency-dependent characteristics of this equivalent impedance, the SSR characteristics could be identified and then compared with field measurements.

2) *Comparing Analysis Results With Field Observations*: Another typical SSR event that occurred on December, 2012 is used to check the effectiveness of model analyses. The system condition (later called Case 1) was as follows: the wind speed was 5.1 m/s and about 90% WTGs were grid-connected. The zoomed-in curve of the captured current at Guyuan substation is plotted in Fig. 18. Using digital Fourier transformation (DFT),

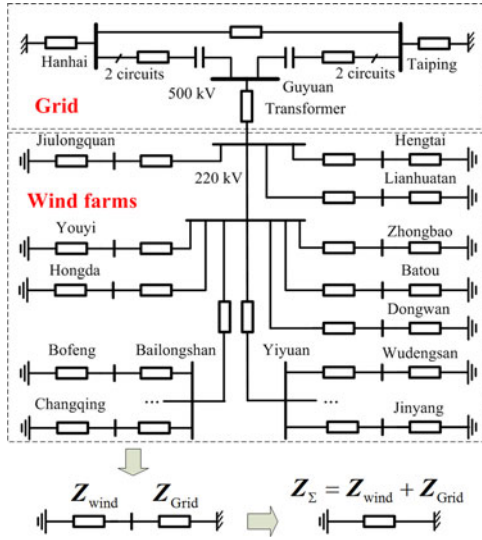


Fig. 17. Impedance network of the Guyuan System.

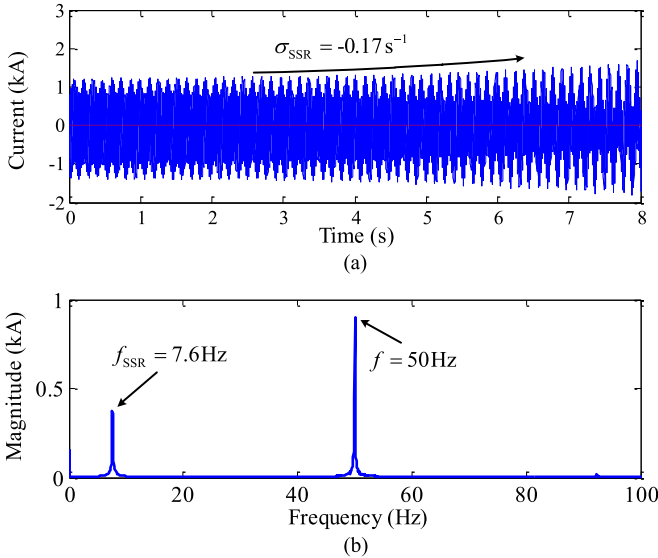
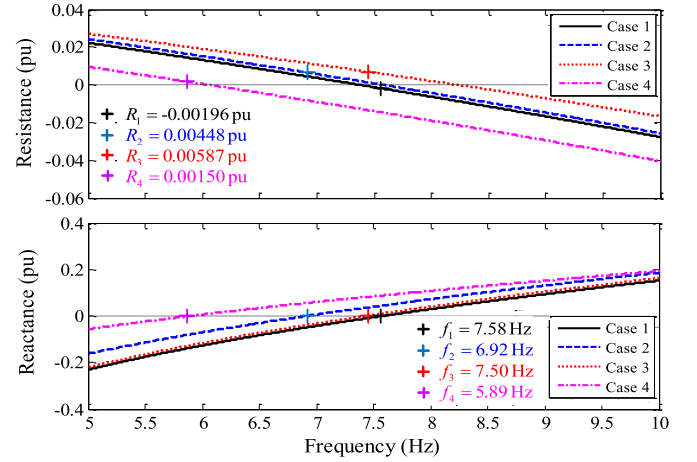


Fig. 18. Field measurements: (a) current at Guyuan. (b) DFT results.

the frequency and damping of SSR are obtained as 7.6 Hz and -0.17 s^{-1} , respectively. The negative damping indicates an unstable SSR. For this case, the equivalent resistance and reactance of the lumped impedance is depicted in Fig. 19. Evidently, there is a zero-crossing point, i.e., $\omega_r = 7.58 \text{ Hz}$, on the reactance-frequency curve, standing for the frequency of the SSR mode. At this frequency point, the equivalent resistance is negative (-0.00196 pu). Further, with the quantitative method proposed in [40], the damping can be worked out to be -0.173 s^{-1} . It is noted that the results of such IM based analyses are consistent with those computed from field measurements.

Besides case 1, Fig. 19 also displays the impedances of other three cases. They are: Case 2 – Switching off the series capacitors of one line connecting Guyuan and Taiping; Case 3 – reducing the proportional gain (K_p) of the RSC by 20%; Case 4 – decreasing the percentage of DFIG from 82.8% to 50%. The

Fig. 19. Frequency-dependent impedance (base value = 0.3174Ω).

zero-crossing frequencies and the corresponding resistances are all listed in the plots.

By comparing the impedances of different cases, we could make the following observations:

Firstly, the existence of all series capacitors is the necessary condition for the SSR incidents, because, otherwise, the equivalent resistance at the oscillation frequency shown as the pointers ‘+’ would be positive, as shown by Case 2 in Fig. 19, indicating a well stabilized subsynchronous mode.

Secondly, the control of DFIG is the primary reason for exciting the SSR due to the fact that it greatly enlarges the negative resistance at the slip frequency, as shown in (2) and confirmed by Figs. 12–14. The curves of Case 3 in Fig. 19 also indicate that a slight (20%) decrease of the proportional gain of the RSC would stabilize the originally unstable SSR.

Thirdly, the number of online WTGs and their allocation among different types also affect the damping (stability) and frequency of SSR. For the Guyuan system, a higher percentage of DFIGs would make the system more vulnerable to SSR, as illustrated by Case 4 in Fig. 19. And the more the online WTGs are, the higher the oscillatory frequency would be, because the reactance of the wind farm become less when more WTGs/IMs are connected in parallel.

Last but not the least, DFIG, PMSG and SEIG have different degrees of participation into this type of SSR. For the discussed Guyuan system, PMSG and SEIG are not so actively engaged. However, such an observation, though predicted by some theoretical analyses and confirmed by the field measurements (see Fig. 10), should be cautiously examined before applying to other WTGs and systems.

V. CONCLUSION

This paper is aimed to fill the gap between theoretical work and practical observations for the newly emerging SSR that is caused by the interaction of DFIGs with series compensation. By analyzing the field data of 58 SSR events captured in the Guyuan system, north China, and checking the observed dynamics with previous theoretical results, the following conclusions can be made:

- 1) This DFIG-associated SSR is a pure electric resonance without the involvement of torsional dynamics. It originates from the L-C resonance but is reinforced by the converter control of DFIG. The latter, by greatly enlarging the negative resistance at the slip frequency, serves as the primary reason for the self-excitation of SSR, even under the context of a very low series compensation. This has been explained by theoretical analysis and confirmed by field observations.
- 2) Theoretically derived and practically measured IMs have indicated that PMSGs are not actively or just passively engaged in the captured SSR incidents. However, such an observation should be cautiously examined before applying to other WTGs and systems, because of the unlimited possibilities of control schemes and parameters.
- 3) Besides the compensation level, wind speed and number of online DFIGs are two important factors that impact the characteristics of SSR because they directly change the IM of a wind farm, especially the negative resistance. In the case of low compensation level, the decrease of wind speed makes the negative slip closer to zero and thus the negative resistance even larger, indicating a more unstable SSR condition.
- 4) In a SSR event, the frequency at different locations is perfectly identical, meaning all WTGs and the network are engaged in the same SSR mode. But its value is not fixed but keeps changing over time as well as the variation of various factors, including network topology, number of online generators, and etc.

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Xiaorong Xie (M'02–SM'14) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1996, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2001. He is currently an Associate Professor in the Department of Electrical Engineering, Tsinghua University. His research interests include flexible ac/dc transmission systems, subsynchronous interaction, and smart grid analysis and control.

Xu Zhang received the B.S., M.S., and Ph.D. degrees from North China Electric Power University, Beijing, China, in 2009, 2012, and 2016, respectively. He is currently working as a Postdoctoral Researcher in the Department of Electrical Engineering, Tsinghua University, Beijing, China. His research interests include analysis and control of subsynchronous interaction between multiple converters and large power system.

Huakun Liu received the B.S. degree from Shandong University, Jinan, China, in 2013. He is currently working toward the Ph.D. degree at Tsinghua University, Beijing, China. His research interests include power system stability analysis and control.

Hui Liu received the Ph.D. degree in electrical engineering from Tianjin University, Tianjin, China, in 2005. He is currently working with the North China Electric Power Research Institute, Beijing, China. His interests include integration of renewables into large power system.

Yunhong Li received the M.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2013. She is currently working with the North China Electric Power Research Institute, Beijing, China. Her interests include analysis and control of power systems with high-penetration of renewables.

Chuanyu Zhang received the B.S. degree from Tsinghua University, Beijing, China, in 2014, where he is currently working toward the M.S. degree. His research interests include power system stability analysis and control.